STORY_DIRECTOR_V4_SOURCE_OF_TRUTH

Story Director V4 — Source of Truth

1. Product goal

Convert a short or long narrative into a complete, director-led, production-feasible, continuity-aware episode plan and video-production pipeline. Each episode lasts 1-4 minutes.

2. Product abstraction

Its hierarchy is:

`Story → Chapter → Episode → Scene → Shot → Episode Video`.

The **Episode Video** is the final viewing unit.

3. Canonical flow

`Upload/Paste → Story Analyzer → SHORT/LONG + CHILDREN/GENERAL + NUMBER OF EPISODES → User confirmation → Visual Style → Director Settings → Story/Director/Character/Location/Style Bibles → Chapter/Episode Plan → Scene Director → Shot Director → Continuity/State Validation → Complete Director Plan → Preview → Explicit approval → Capability/Feasibility Gate → Provider Prompt Compiler → Clip → Validation/targeted repair → Deterministic assembly → Final validation → Watch`.

Scene planning and Shot planning are distinct responsibilities even if an implementation temporarily batches calls. The canonical representation must preserve the distinction..

4. Independent axes

Source Scale: `SHORT | LONG`. Long stories use Chapter grouping; short stories may hide meaningless Chapter hierarchy.

Story Mode: CHILDREN | GENERAL.

CHILDREN prioritizes clarity, visual readability, emotional safety, simple progression, recognisability, moderate pacing, shorter dialogue, and lower Shot complexity. Direction should remain easy to follow, emotionally legible, visually expressive, and appropriate for younger audiences.

GENERAL targets the viewing experience and directing grammar of **high-retention, highly engaging short-form drama**. It should not merely permit greater narrative or cinematic complexity; the Director should actively shape Episodes, Scenes, and Shots for sustained viewer attention, emotional momentum, and continued-viewing desire while preserving the source story, canonical facts, character logic, and continuity.

GENERAL direction should therefore favor:

- **Fast narrative entry** — establish the immediate situation, desire, conflict, anomaly, or unanswered question quickly rather than spending excessive runtime on exposition.
- **High information and event density** — each Scene should materially advance plot, character state, conflict, revelation, or anticipation; avoid visually polished but narratively idle material.
- **Frequent engagement beats** — create meaningful micro-payoffs, discoveries, reversals, escalations, emotional turns, reveals, or new questions throughout an Episode rather than relying on a single climax.
- **Escalating tension and stakes** — conflict should develop rather than remain static; successive beats should increase pressure, alter expectations, complicate goals, or deepen emotional investment.

- **Strong emotional contrast** — use changes in power, expectation, relationship, danger, hope, embarrassment, satisfaction, surprise, or other story-supported emotions to prevent a flat emotional trajectory.
- **Setup → pressure → payoff loops** — establish expectations or unresolved tensions, develop them efficiently, and provide satisfying payoffs while opening the next meaningful narrative question.
- **Open loops and curiosity** — strategically preserve unanswered questions, incomplete information, impending consequences, or anticipated confrontations that create a reason to continue watching.
- **Strong Scene transitions** — enter Scenes as late as comprehension allows and leave once their dramatic function has landed, avoiding unnecessary arrival, departure, repetition, or connective footage.
- **Purposeful cinematic variation** — Shot scale, composition, camera movement, reaction Shots, inserts, pacing, and visual emphasis should serve the current dramatic beat rather than vary arbitrarily.
- **Reaction and consequence visibility** — important actions, revelations, and dialogue should receive sufficient visual or emotional reaction so that their dramatic impact is felt rather than merely stated.
- **Episode-level hooks and payoffs** — each Episode should function as a satisfying viewing unit with its own progression and payoff while, when supported by the story, ending with a strong unresolved development, reversal, revelation, decision, threat, anticipation beat, or other continuation hook.

These are **directing principles, not mandatory tropes or mechanical quotas**. The system must not invent unsupported melodrama, conflict, cliffhangers, reversals, dialogue, or character behavior merely to increase engagement. It should first discover the strongest hooks, tensions, emotions, mysteries, reversals, and payoffs already latent in the source, then amplify them through **selection, ordering, pacing, emphasis, cinematography, performance, and Episode/Scene/Shot construction**.

GENERAL may use denser dialogue, subtler narrative, broader cinematography, parallel action, and greater Shot complexity than CHILDREN, but complexity is justified only when it improves storytelling or engagement.

Story Mode and Visual Style remain independent. Story Mode governs narrative and directing behavior; Visual Style governs the audiovisual rendering language. A GENERAL story may therefore be realistic, animated, stylized, cinematic, illustrated, or otherwise visually rendered without changing its underlying high-retention directing principles.

Story Mode and Visual Style are independent.

5. Hierarchy

- **Chapter:** long-form narrative grouping; may but need not match source chapters mechanically.
- **Episode:** final viewing unit; ~3–5 minutes is a directing target, not a hard fixed duration.
- **Scene:** narrative/cinematic unit; not assumed to equal one model call.
- **Shot:** actual video-generation unit; one Scene may contain multiple Shots.
- **Episode Video:** assembled validated viewing artifact.

Episode count is Director-owned by default. Advanced settings may provide an approximate target. Scene/Shot counts are Director-owned.

6. Director responsibility

Interpret major characters, central conflict, themes, narrative arc, emotional beats, turning points, climax/resolution and Chapter/Episode structure. Scene-level directing owns dramatic objective, scene purpose, emotional progression, geography/blocking basis, lighting logic, active production design, atmosphere, sound intent and beginning/end state. Shot-level directing owns coverage, Shot purpose, audience-information change, framing, cinematography, performance/micro-action, blocking application, visible environment, lighting application, timing and edit relationship.

The authoritative professional contract is `V4_PROFESSIONAL_DIRECTING_SYSTEM.md`. The expanded DOCX is reference material, not a competing implementation authority.

7. Normal user settings

Story; Story Mode confirmation; Visual Style; preferred Episode length (2/3/4/~5 min); Faithful/Balanced/Creative policy; voice preference where supported; optional additional direction. Do not require Scene/Shot counts.

8. Canonical Bibles

- **Story Bible:** source facts/events with stable IDs, separate from adaptation decisions.
- **Director Bible:** pacing, dialogue density, cinematography, fidelity, episode-duration target, transitions, age/safety and production assumptions.
- **Character Bible:** source facts, relationships, detailed appearance (facial charactors, skin color, hair style, height, etc), clothing, personality, acting direction, canonical references and voice identity.
- **Location Bible:** identity, architecture, components in the background/surroundings/weather etc, props, palette, lighting, time variants.
- **Voice Bible:** age presentation, tone, energy, pace, explicitly selected accent and emotional range.
- **Video Style Bible:** medium, texture, palette, lighting, composition plus Motion Language: camera energy, character/environment motion, transitions and avoid rules.

Initial presets: Watercolor Storybook; Gouache Storybook; Paper Cutout; Ink Wash; Graphic Novel; Cinematic Painterly; Stylized 3D; Anime-inspired Cinematic.

9. Pre-production references

Retain MFLUX/Z-Image but repurpose it from final-output generator to character portraits/full-body/expression references, location references and keyframes/visual references. Persist prompt, provider/model, attempt, asset, validation and lineage.

10. Provider-aware, provider-independent planning

Maintain a Provider Capability Registry: clip duration, resolution/aspect ratio, reference-image support, start/end frame, native audio, dialogue/speech, reproducibility, rate/concurrency, cost/usage and restrictions. Canonical domain entities remain provider-independent.

11. Canonical Scene and Shot Specs

The canonical Scene and Shot representations are structured filmmaking contracts, not only prose containers.

A Scene minimally carries: identity/lineage; purpose; dramatic objective/conflict; source events; location/characters; narrative and emotional beats; information/reveal intent; beginning/end state; scene geography and blocking basis; lighting/environment/production-design intent; dialogue/sound intent; continuity requirements; duration; and child Shots.

A Shot minimally carries: identity/lineage; purpose/role; director/narrative focus; information change; location/characters; subject/visual focus; exact visible action; structured performance; structured cinematography; blocking; visible environment/production design; lighting; timing; sound/audio intent; editing relationship; start/end continuity state or state delta; immutable locks; allowed changes; references; negative constraints; style/motion language; feasibility; and provider prompt lineage.

Provider prompts compile deterministically from this richer representation. Provider prompt text is downstream execution data and must not replace canonical directing state.

12. Feasibility gate

Every Shot passes deterministic feasibility validation before a Video Provider call. Technical capability constraints must not silently invent filmmaking decisions. If a planned Shot exceeds provider capability, route it back through meaningful re-direction/decomposition (or a deterministic split only when semantics are provably preserved), then revalidate. Never duplicate the same action and label the remainder generic “reaction coverage” merely to satisfy duration.

12A. Automated double verification and non-blocking self-healing

Verification is a staged quality gate, not a permanent stop condition. Gate A verifies structured plans before expensive generation using deterministic checks plus narrow dramaturgy/film-direction/continuity critics. Gate B verifies the rendered Episode using technical checks, sampled intra-Shot frames, adjacent-Shot end/start comparisons, visual continuity/causality criticism and Episode-level filmmaking review.

The canonical contract is `V4_AUTOMATED_VERIFICATION_SYSTEM.md`. Findings route to the smallest owner (`PLAN | COMPILER | PROVIDER | ASSEMBLY | UNCERTAIN`), trigger bounded automatic repair/regeneration when safe, and reverify automatically. Low-confidence warnings do not block human handoff. When budgets are exhausted or evidence is uncertain, transition to `REVIEW_REQUIRED`; never remain indefinitely blocked in verification.

Rendered checks cover character identity/proportion/scale, style, location/geography/landmarks, story-relevant prop identity/topology/state, action readability, contact plausibility, physical causality, emotional/performance continuity, Shot-purpose fulfillment, reaction/reveal readability, time/light/weather, intra-Shot coherence, inter-Shot continuity and state-transition plausibility.

12B. Model routing and BYOK

The system supports a practical local default and optional stronger local/API models. Normal users need not choose models; `Settings → Advanced → AI Models` provides global presets and optional per-stage routing for analysis, Scene Director, Shot Director, critics and repair. Hosted models use user-provided API credentials stored outside Project artifacts/logs/state. Model routing is capability-based through a Model Capability Registry, including text/image/multi-image/video and structured-output capabilities. A missing visual model marks visual checks `UNVERIFIABLE` and proceeds to human review rather than blocking. See `V4_MODEL_ROUTING_AND_BYOK_POLICY.md`.

13. Approval boundary

All expensive generation requires complete relevant planning, Plan Preview and explicit approval. Preview exposes story interpretation, Chapters/Episodes, summaries/goals, canonical characters/locations, style, duration estimate, production requirements, warnings and provider availability.

14. Local rendering policy

Local generation has no per-project Episode count limit. Complete story → complete plan → all Chapters/Episodes inspectable and eligible for local production when compatible local video and voice providers are configured. Execute heavy local work serially and retain provider feasibility, resource safety, approval, persistence, validation, repair and device-failure boundaries. An Episode is stopped only when its actual plan, provider or device cannot support it; Episode numbering never selects an external provider. This policy reflects the product owner's later demo decision to permit as many locally feasible Episodes as the device can undertake.

15. External provider state

External video, voice and model APIs are optional selections for larger or better models. They are never selected because an Episode is numbered 2 or higher. Without a compatible configured local provider, Episodes remain planned and inspectable and show provider unavailability; no paid fallback occurs. External selection retains explicit credentials, capability checks, approval and call budgets.

16. Credentials

Credentials never enter Story State, Bibles, prompts, artifacts, logs, evaluation exports or lineage. Secrets live only in the provider configuration boundary.

17. Validation

Deterministic: schemas, IDs, references, lineage, source-event coverage/order, duration arithmetic, provider feasibility, assets, resource policy, state transitions, assembly/timeline integrity and credential absence.

Semantic: source fidelity, coherence, character/location continuity, Story Mode suitability, style/motion consistency, dialogue/action alignment, Shot intent vs Clip, and cross-Episode continuity.

18. Targeted repair

`Artifact → Validate → Diagnose → repair smallest affected artifact → revalidate`. Never regenerate unrelated successful work. Preserve failed attempts/evidence. Repair is bounded; exhaustion becomes `REVIEW_REQUIRED`. Rebuild downstream artifacts selectively by lineage.

19. Assembly

AI plans; deterministic code assembles. Timeline/order/duration and artifact references must be reproducible from persisted production state.

20. Persistence/resume/lineage

Preserve V3.1's persisted pipeline strengths. Every artifact records stable identity, parent/source inputs, version, status and dependency lineage. A fresh process must reload planning/completed/interrupted runs without loading heavy models until needed.

21. Canonical UI

Only five top-level areas: **Create, Director Plan, Production, Watch, Evaluate**. Debug details live under Production details.

22. Watch

Watch is Player + Episode Library. Chapter is first-class navigation for long stories. Episode rows show narrative title, duration and restrained production state. Hide model, prompt count, tokens, validation score and attempts in normal viewing.

23. UI language

Avoid implementation buzzwords such as “Director Agent is thinking.” Prefer “Planning story structure…”, “Designing Episode 1…”, “Checking visual continuity…”. Use restrained status color.

24. Selective invalidation

Style change preserves source/story facts and identities but rebuilds style/visual references/prompts/rendered assets. Story Mode change invalidates Director planning downstream while preserving source facts. Episode-duration change reconsiders Episode boundaries, Scene allocation and Shot pacing.

25. Interactions

V3.1 interactions remain in the frozen baseline but are removed from the canonical V4 pipeline. V4 is linear story-to-video adaptation.

26. Evaluation

Separate **planning quality** from **rendered-video quality** and compare simpler baselines without assuming V4 wins. Never edit frozen cases to improve results. Report quality, reliability, calls, latency, cost and repair behavior.

27. Non-negotiables

1. Episode Video is final viewing unit.
2. Scene is narrative unit; Shot is generation unit.
3. Chapter may be hidden for short stories.
4. Complete planning is not limited by local rendering.
5. Local rendering has no arbitrary Episode count cap; provider and device feasibility still govern each Episode.
6. Episode 2+ stays fully planned/inspectable.
7. Story Mode and Visual Style are decoupled.
8. Style Bible includes Motion Language.
9. Character/Location/Voice identities are canonical.
10. Existing image pipeline is retained for references.

11. Provider capabilities are explicit.
12. Scene and Shot are separate.
13. Every Shot passes feasibility validation.
14. Plan Preview + approval precede expensive generation.
15. Repair is smallest-affected-artifact and bounded.
16. Unreliable repair becomes REVIEW_REQUIRED.
17. Assembly is deterministic where possible.
18. Watch hides production complexity; Production exposes it progressively.
19. V3.1 frozen implementation/evaluation evidence is preserved.
20. V4 must not claim superiority before comparative evidence exists.
21. Scene Director and Shot Director responsibilities are explicitly separated.
22. Professional directing decisions that require validation/reuse are structured fields, not only free text.
23. Continuity is state transition, not merely prose reminders.
24. Provider limitations cannot silently rewrite directing intent.
25. Rich Director representation and compact provider instruction remain separate layers.